

According to these rules, it's easy to see that the remaining 8 stones will be removed in 4 moves.

The 11th stone is removed in the 5th move by the 1st player (5 is odd).

∴ We have shown that there always exists a winning strategy for the 1st player. $\square$

Adapting Existing Proofs

An excellent way to look for possible approaches that can be used to prove a statement is to take advantage of existing proofs of similar results.

Example 16: In Example 11 of Section 1.7 we proved that $\sqrt{2}$ is irrational.

We now conjecture that $\sqrt{3}$ is irrational. Can we adapt the proof in Example 11 in Section 1.7 to show that $\sqrt{3}$ is irrational?

Solution: To adapt the proof in Example 11 in Section 1.7, we begin by mimicking the steps in that proof, but with $\sqrt{2}$ replaced with $\sqrt{3}$. First we suppose that $\sqrt{3} = c/d$ where the fraction c/d is in lowest terms. Squaring both sides tells us that $3 = c^2/d^2 \Rightarrow c^2 = 3d^2$.

a) Can we show that 3 must be a factor of both c and d , similar to how we used the ~~lemma~~ $2b^2 = a^2$ in Example 11 in Section 1.7 to show that 2 must be a factor of both a and b ? (Recall that an integer s is a factor of the integer t if t/s is an integer. An integer n is even if and only if 2 is a factor of n .)

It turns out that we can, but we need some ammunition from number theory, which we will develop in Chapter 4. We sketch out the remainder of the proof, but leave the justification of these steps until Chapter 4.

Because 3 is a factor of c^2 , it must also be a factor of c . Furthermore, because 3 is a factor of c , 9 is a factor of c^2 , which means that 9 is a factor of $3d^2$. This implies that 3 is a factor of d^2 , which means that 3 is a factor of d . This makes 3 a factor of both c and d , which contradicts the assumption that c/d is in lowest terms.